

# Paul (Chong Wa) Lai

Postdoctoral Researcher in Astrophysics

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## Research Interests

- Studying the cosmic-ray and gas distributions in the Milky Way with multi-messenger signals
- Probing astrophysical magnetic fields with polarisation and Faraday rotation
- Polarised radiative transfer in large-scale magnetised diffused media

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## Employment

From 2026 Sep    **Postdoctoral researcher** at RWTH Aachen University

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## Education

- 2022 – 2026    **Ph.D. in Astrophysics**  
University College London, Mullard Space Science Laboratory,  
Department of Space and Climate Physics  
*Thesis:* Neutrinos and gamma-rays from the Milky Way Centre  
*Supervisors:* Prof. Kinwah Wu, Dr. Matteo Agostini, and Prof. Foteini Oikonomou
- 2019 – 2022    **M.Phil. in Astrophysics**  
The University of Hong Kong, Department of Physics  
*Thesis:* Radio polarization study of the young pulsar wind nebula in supernova remnant G21.5–0.9  
*Supervisor:* Dr. Stephen Chi-Yung Ng
- 2015 – 2019    **B.Sc. (1st class honours) in Theoretical Physics**  
The Chinese University of Hong Kong, Department of Physics  
*Thesis:* Behaviour of photons in modulating spacetime  
*Supervisors:* Dr. Po Kin Leung and Prof. Kinwah Wu

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## Publications (\*highlighted publications)

6. J. Y. H. Chan, A. Y. L. On, **P. C. W. Lai**, and Kinwah Wu, “Bridging Theory and Observation in the SKA Era: A Cosmological Polarized Radiative Transfer Framework for Point-to-Point Polarized Sky Comparisons”, Chapter in *Advancing Astrophysics with SKA II* (2026)
- 5\*. **P. C. W. Lai**, Beatrice Crudele, Matteo Agostini, Hayden P. H. Ng, Ellis R. Owen et al., “Neutrinos as a New Tool to Characterise the Milky Way Centre”, [PRD Letter, 113, L061306 \(2026\)](#)
4. **P. C. W. Lai**, Chi-Yung Ng, Shumeng Zhang, “High Resolution VLA Radio Observations of the Boomerang Pulsar Wind Nebula”, submitted to *ApJ*, [arXiv:2602.20230](#)
3. Alvina Y. L. On, Jennifer Y. H. Chan, **P. C. W. Lai** and Kinwah Wu, submitted to *MNRAS*, “Effects on the polarisation properties of ensembles of point sources embedded in an extended diffuse media”.
- 2\*. **P. C. W. Lai**, Kaye J. Li, Y. X. Jane Yap, Kinwah Wu and Albert K. H. Kong, “Spectropolarimetry of Synchrotron Radiation from Relativistic Electrons with Anisotropic Pitch-angle and Various Energy Distributions”, [MNRAS, Volume 542, Issue 2, Pages 902–916 \(2025\)](#)
- 1\*. **P. C. W. Lai**, Chi-Yung Ng, N. Bucciantini, “Magnetic Field Structure and Faraday Rotation of the Plerionic Supernova Remnant G21.5–0.9” [ApJ, 930, 1 \(2022\)](#)

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## Conference Proceedings

2. **P. C. W. Lai**, B. Crudele, M. Agostini et al., “Neutrinos as a new tracer of the gas distribution in the Milky Way Centre”. [39th International Cosmic Ray Conference \(ICRC\), id. 1089 \(2025\)](#)
1. **P. C. W. Lai**, M. Agostini, F. Oikonomou et al., “Neutrino Imaging of the Galactic Centre and Millisecond Pulsar Population”, [38th International Cosmic Ray Conference \(ICRC\), id. 1069 \(2023\)](#)

## Grants, Scholarships, Studentships and Awards

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|-------------|--------------------------------------------------------------------------------------------------------------------------|
| 2025        | <b>RAS Grant – Conference Travel Subsistence (£1,000)</b> , Royal Astronomical Society                                   |
| 2025        | <b>Research Student Conference Fund (£300)</b> , Institute of Physics (IOP)                                              |
| 2025        | <b>MSSL Travel Grant (£500)</b> , Mullard Space Science Laboratory, University College London                            |
| 2024        | <b>Cosmoparticle Initiative Travel Grant (£2,000)</b> , University College London                                        |
| 2023        | <b>MSSL Travel Grant (£200)</b> , Mullard Space Science Laboratory, University College London                            |
| 2022 – 2025 | <b>Graduate and Overseas Research Scholarship</b> , University College London                                            |
| 2019 – 2021 | <b>Postgraduate Scholarship</b> , Graduate School, The University of Hong Kong                                           |
| 2017 – 2019 | <b>The Dean's Honours List (3 times)</b> , Faculty of Science, The Chinese University of Hong Kong                       |
| 2019        | <b>C N Yang Prize</b> , Department of Physics, The Chinese University of Hong Kong                                       |
| 2018        | <b>UCL MAPS Dean's Summer Student Scholarship</b> , Faculty of Mathematics & Physical Science, University College London |
| 2018        | <b>C N Yang Scholarship for Research Experience</b> , Department of Physics, The Chinese University of Hong Kong         |
| 2017        | <b>Physics Prize</b> , Department of Physics, The Chinese University of Hong Kong                                        |

## Invited Talks

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| 2026 Mar 4     | <b>SPIMAX Seminar Series</b> , Oxford University, UK                                                                                                                                |
| 2025 Aug 22    | <b>Seminar</b> , Astronomical Institute, Tohoku University                                                                                                                          |
| 2024 Jun 25    | <b>Seminar</b> , Norwegian University of Science and Technology, Trondheim, Norway                                                                                                  |
| 2023 Dec 19–20 | <b>Invited Talk</b> , the Energetic Universe and Multi-Messengers Workshop at National Taiwan Normal University                                                                     |
| 2023 Dec 15–17 | <b>Invited Talk</b> , National Centre Theoretical Sciences (NCTS) Annual Theory Meeting: Particle Physics, String Theory, Cosmology, and Astrophysics at National Taiwan University |
| 2023 Jul 25    | <b>Seminar</b> , Department of Earth and Space Science, The University of Osaka                                                                                                     |

## Conference and Workshop Presentations

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| 2025 Jul 14-24    | <b>Contributed Talk</b> , International Cosmic Ray Conference (ICRC 2025) in Geneva, Switzerland |
| 2025 Jul 7-11     | <b>Contributed Talk</b> , National Astronomical Meeting (NAM), Durham, UK                        |
| 2024 Dec 9-13     | <b>Talk and Poster</b> , Cosmic Rays and Neutrinos in the Multi-Messenger Era, Paris, France     |
| 2023 Oct 4–6      | <b>Contributed Talk</b> , IV Gravi-Gamma-Nu Workshop in L'Aquila, Italy                          |
| 2023 Jul 26–Aug 3 | <b>Poster Presentation</b> , International Cosmic Ray Conference (ICRC 2023) in Nagoya, Japan    |

## Teaching and Outreach Experience

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| 2026 | <b>Sharing session speaker</b> , Astronomical Training Programme for Secondary Students, Hong Kong                                                                                                |
| 2025 | <b>Teaching Assistant of High Energy Astrophysics (SPCE0012)</b> , Department of Space and Climate Physics, University College London                                                             |
| 2024 | <b>Teaching Assistant of High Energy Astrophysics (SPCE0012)</b> , Department of Space and Climate Physics, University College London                                                             |
| 2024 | <b>Project Instructor</b> (teaching high school students how to compose RGB images of astrophysical objects using real telescope data), MSSL Work Experience Week 2024, University College London |
| 2021 | <b>Teaching Assistant of Advanced Astrophysics (PHYS4656)</b> , Department of Physics, The University of Hong Kong                                                                                |
| 2021 | <b>Chief Laboratory Demonstrator</b> , Department of Physics, The University of Hong Kong                                                                                                         |
| 2020 | <b>Teaching Assistant of Astronomy Laboratory (PHYS3660)</b> , Department of Physics, The University of Hong Kong                                                                                 |
| 2020 | <b>Teaching Assistant of Introductory Electricity and Magnetism (PHYS2255)</b> , Department of Physics, The University of Hong Kong                                                               |
| 2020 | <b>Certificate in Teaching and Learning in Higher Education</b> , Centre for the Enhancement of Teaching and Learning, The University of Hong Kong                                                |

## Awarded Telescope Proposals

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- 2021 **Principal Investigator**, the Very Large Array (VLA)  
*Co-I*: Ruolan Jin and Chi-Yung Ng  
*Project 21A-058*: high resolution radio study of the Boomerang pulsar wind nebula (1.5 hr)
- 2021 **Co-Investigator**, the Very Large Array (VLA)  
*PI*: Ruolan Jin, *Co-I*: Chi-Yung Ng and **Paul Chong Wa Lai**  
*Project 21A-265*: probing the youngest pulsar wind nebula in the galaxy (1.5 hr)

## Scientific Service

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- Since 2026 **Referee** for American Astronomical Society (AAS) journals
- 2023 – 2024 **Seminar Organiser**, Department of Space and Climate Physics, University College London

## Software/Code Experience

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**CPRT** (polarised radiative transfer), **DustEM** (interstellar dust emission), **AM3** (lepto-hadronic interactions), **CASA** (radio data reduction tool)

## Languages

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English (Proficient), Mandarin (Fluent), Cantonese (Native)